



Pyrolysis study of 1,2,4-trimethylcyclohexane with SVUV-photoionization molecular-beam mass spectrometry

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ABSTRACT

Pyrolysis of 1,2,4-trimethylcyclohexane (T124MCH) at 30 and 760 Torr has been investigated by using VUV-synchrotron photoionization molecular-beam mass spectrometry. In general, the ambient pressure leads to peak-shaped profiles of light hydrocarbons, while the monotonically increasing curves are observed for light hydrocarbons and aromatics in pyrolysis of T124MCH at low pressure. A detailed mechanism involving 530 species and 3160 reactions developed by the authors previously was used for the simulation of T124MCH with reasonable predictions. The consumption of T124MCH followed similar patterns at both pressures, namely the H-abstractions on the first, secondary, and tertiary carbon atoms are the main consumption reactions of T124MCH. The isomerization reactions transform the C₉H₁₇ radicals to branch-chain hydrocarbons, which are the precursors of light hydrocarbons such as ethane and propene. Sensitivity analysis indicates that the CH₃ consumption reactions are the most significant promoting reactions at both pressures. Isomerization reactions of T124MCH tend to play significant inhibiting effect on T124MCH pyrolysis. Compared to T124MCH oxidation, T124MCH pyrolysis tends to produce large quantities of benzene and toluene. These results will improve the understanding of the combustion and soot formation of T124MCH as a potential aviation surrogate fuel.

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1. Introduction

In recent years, the combustion studies of aviation fuels, as the energy source of aircrafts, have attracted more and more attention with respect to high efficiency and low pollutant emissions. Zheng et al. [1] identified the compositions of RP-3 aviation kerosene with alkanes (53.0%), cycloalkanes (37.7%) and minor quantity of aromatics as well as olefins with gas chromatography-mass spectrometer (GC-MS). Widegren and Bruno. [2] presented the compositions of Jet-A POSF 4658: alkanes (68.2%), aromatics (25.5%), naphthalenes (3.0%) and cycloalkanes (3.3%) with the same technology. Surrogate fuels, which make the modeling the behavior of real aviation fuels feasible, have been widely used in the combustion field. Recently, cyclohexane (CH) and alkyl cyclohexanes have attracted more attention as the important components of surrogate

model fuels, as they can generate both alkanes and olefins in the combustion process [3–6].

Most of the cyclohexane combustion studies were focused on CH and alkyl cyclohexanes, especially methylcyclohexane (MCH). Several studies of MCH were reported with respect to its pyrolysis [7–9], flame [10,11], oxidation [12], and ignition [13,14]. Long side-chains alkyl cyclohexane (e.g., ethylcyclohexane, propylcyclohexane, butylcyclohexane and etc.) were also reported regarding their combustion experiments and kinetic modeling. However, the combustion studies of alkyl CH with multiple side-chains are quite limited. Gillespie et al. studied the pyrolysis of 1,4-MCH and identified a number of products by using a flow reactor in 1979 [15]. Kang et al. measured the ignition delay time of 1,2-MCH and 1,3-MCH in a motored engine in 2015 [16]. Eldeeb et al. studied the ignition and pyrolysis of 1,3-MCH under high-temperature and presented a detailed kinetic model [17]. Up to 2018, there were neither experimental nor modeling studies on trimethylcyclohexane (tri-MCH) available in the literature. Tri-MCH can play an important role in kerosene surrogate, which is a typical kind of cycloalkanes and could isomerize to more kinds of linear alkenes than ethyl-CH or n-propylcyclohexane (NPCH) which are widely

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used in surrogate fuels. More recently, our group reported an oxidation study on 1,2,4-trimethylcyclohexane (T124MCH) with a new kinetic model [18]. However, due to the lack of experimental data, the model still needs to be further validated against other kinds of experimental results such as pyrolysis data.

This work aims to identify and quantify intermediates and products in the pyrolysis of T124MCH. Based on the experimental data, the previous kinetic model which can reproduce the experimental results in the oxidation work would be validated. Rate-of-production (ROP) and sensitivity analyses were performed to identify the consumption and sensitive pathways of T124MCH pyrolysis. The validated mechanism could provide a detailed study on tri-MCH as a potential surrogate compound, and improve the understanding of its combustion characteristics of diesel and jet fuels for future applications.

2. Experimental

The experiments were carried out at the National Synchrotron Radiation Laboratory (NSRL) in Hefei, China, which has been described elsewhere [19], and only a brief description is given here. A tubular flow reactor made of corundum ($\alpha\text{-Al}_2\text{O}_3$) whose outer diameter, inner diameter, and length is 10, 7 and 220 mm respectively, was used. T124MCH was vaporized by a high-pressure syringe pump, and carried by helium. The total flow rate was 1000 sccm, consisting of 985 sccm of He, 10 sccm of Kr for calibration, and 5 sccm of T124MCHs. The flow rates were regulated by MKS mass flow controllers. During the experiments, the pyrolysis temperature was measured with an S-type thermocouple. The pyrolysis products were sampled by a small quartz nozzle and ionized by SVUV photo-ionization mass spectrometry. The results of ionization were collected by data acquisition software for subsequent analysis.

The pyrolysis experiments included two parts. The first part was to obtain mass spectra at different temperatures and get mole fraction-temperature profiles of every species. T124MCH was scanned with ionization energies of 15.34, 14.50, 13.00, 11.00, 10.50, 10.00, 9.50 and 9.00 eV, temperature ranges of 875–1200 K at 30 Torr and 775–1100 K at 760 Torr. The other part was the mass spectra plot of the mixture at different ionization energies with the scanning range of 8.50–15.34 eV, using for quantitative analysis of unknown products in pyrolysis process. In order to detect pyrolysis products as more as possible, the scanning temperature was 1100 K at 30 Torr and 1000 K at 760 Torr. The accuracy of known components (detailed ionization data can be seen in [20]) is 25%. The mole fractions were obtained through data analysis and calculation for unknown components with an error factor of 2. The measured data are available in Tables S1 and S2 in Supplemental Material (SM).

3. Modeling

The PFR code in the CHEMKIN-II software was used to simulate the pyrolysis experiments. The details of kinetic model were presented in our previous work [19]. This is a high-temperature oxidation mechanism coupled with an acetylene oxidation mechanism [21]. The newly developed T124MCH sub-mechanism was estimated based on the previous mechanism of MCH proposed by Wang et al. [22]. According to Wang's mechanism, the initiation of MCH gives rise to five C_7H_{13} radicals, which then would decompose to methylcyclohexene or isomerized to linear C_7 hydrocarbons. It is reasonable to believe that cyclic C_9H_{17} radicals, derived from T124MCH, will generate linear C_9H_{17} radicals or trimethylcyclohexenes. The C_9 hydrocarbons will decompose to light hydrocarbons such as C_2H_4 or C_3H_6 via β -scission, while the trimethylcyclohexenes will continue to consume through the H-abstractions

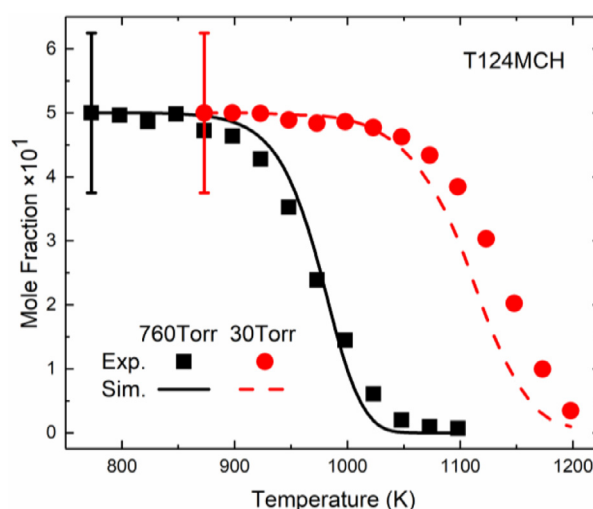


Fig. 1. Experimental and modeling (kinetic model comes from [19], similarly hereinafter) results of 1,2,4-trimethylcyclohexane pyrolysis at 760 and 30 Torr.

to form 1,2,4-trimethylbenzene (T124MB). The consumption pathways from T124MB to xylenes, toluene and other aromatics were also considered [23].

4. Results and discussion

4.1. Mole fraction profiles

More than 44 species were detected in the pyrolysis of T124MCH, as shown in Table S3 (see SM). Generally, ethylene, propylene, allene, propyne, butadiene and butyne are main light hydrocarbons formed in pyrolysis at both pressures, among which ethylene plays a dominant role. Benzene, toluene and ethylbenzene were observed at both pressure, and benzene is the most abundant among aromatics. At atmospheric pressure, T124MCH begins to decompose at a lower temperature with more ethylene, allene, benzene, toluene and ethylbenzene. More propylene, propyne and 1,3-butadiene are formed at low pressure. The species profiles are peak-shaped curve at 760 Torr, while the species increase monotonically at 30 Torr. To further reveal the formation pathways of intermediates, kinetic modeling was conducted with the previously reported model [19].

The mole fraction profiles of T124MCH under both pressures are shown in Fig. 1. The current mechanism could well predict the mole fractions and tendencies along the whole temperature range. Consumption of T124MCH occurred at 875 K in 760 Torr. This value shifted to 1050 K at the pressure of 30 Torr. The temperature difference of initial consumption under two pressures was 175 K, indicating that T124MCH started to decompose at a lower temperature under higher temperature. This value maintained during subsequent reactions. T124MCH was consumed completely at 1075 K under 760 Torr while less than 10% of T124MCH still remained when the pressure dropped to 30 Torr. Simulation could predict the pressure effect, suggesting that the detailed chemical kinetic model of T124MCH developed in [19] could be applied to predict both oxidation and pyrolysis of T124MCH.

The mole fraction profiles of several main light hydrocarbon intermediates during T124MCH pyrolysis are displayed in Fig. 2. Ethylene is the most abundant intermediate during the pyrolysis of T124MCH at 760 Torr. This phenomenon is quite similar to the previous results of oxidation process at the equivalence ratio of 2.0. The difference between pyrolysis and oxidation results in [19] is that 1,3-butadiene was not abundant under the oxidation condition. T124MCH tends to decompose by reactions with various

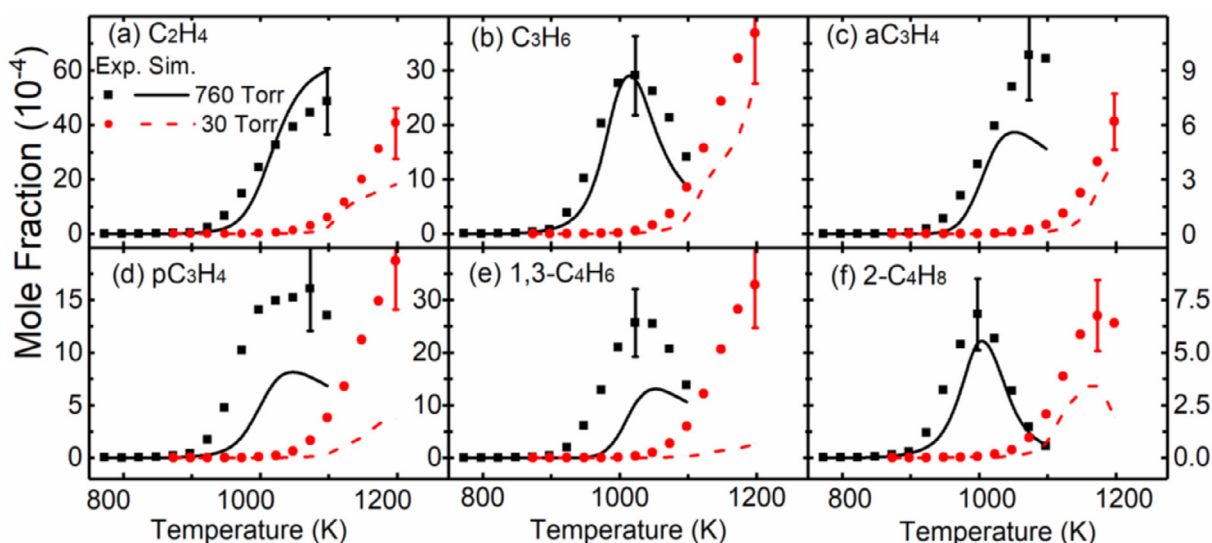


Fig. 2. Experimental and modeling results of light hydrocarbons during pyrolysis of T124MCH.

oxygenated radicals after ring opening isomerization under oxidation conditions, which results in the formation of abundant methane, ethylene and propylene. Minor T124MCH will dehydrogenate and form 1,3-butadiene and aromatic hydrocarbons during the oxidation process. Ethylene and propylene could be formed by the β scission of the radicals via the ring opening and isomerization of T124MCH. Moreover, allene, propyne and 2-butylene could be produced by similar β scissions, but much less radicals are involved, which leads to the fact that allene, propyne and 2-butylene exhibit mole fraction profiles similar to those of ethylene and propylene with lower values. The formation routes changed when pressure dropped to 30 Torr. The formation temperatures of the 6 intermediates at 760 Torr are lower than the corresponding ones at 30 Torr. The mole fraction values of the six intermediates differed from each other, but the peak values of ethylene and 2-butylene approach to each other. Butadiene is less formed at 30 Torr than 760 Torr. However, propyne, 1,3-butadiene and 2-butene have more yields at lower pressure. As a dominant product, the production of ethylene is less affected by pressure since its formation pathways mainly through the ring-opening radicals can compensate with each other. 2-Butylene, with similar tendency of ethylene, is produced after two or more β scission reactions, which accounts for less formation of 2-butylene than ethylene. Propylene, allene and propyne are mainly formed by β -scissions with branched chains, and the differences among the three species demonstrate that the branched chains have obvious influence on the pressure effect.

The mole fraction profiles of aromatic hydrocarbons are presented in Fig. 3. Benzene is the most abundant aromatic intermediate, whose peak value equals to 5 and 10 times of peak values of toluene and ethylbenzene, respectively. Benzene could be formed by hydrogen and methyl group abstraction alternatively from T124MCH. During the formation of benzene ring, the poly-substituted cyclohexanes tend to lose methyl groups directly instead of dehydrogenation at a substituent. According to ROP analysis, the formation of ethylbenzene relies mainly on toluene. In general, the simulated results are in reasonable agreement with the measured profiles, while benzene and toluene are slightly over-predicted and ethylbenzene is under-predicted. Two reasons could account for the differences between the simulated and experimental results. On the one hand, the formation of aromatic hydrocarbons generated from alkylcyclohexanes will experience more steps than the formation of light hydrocarbons. Some reactions were

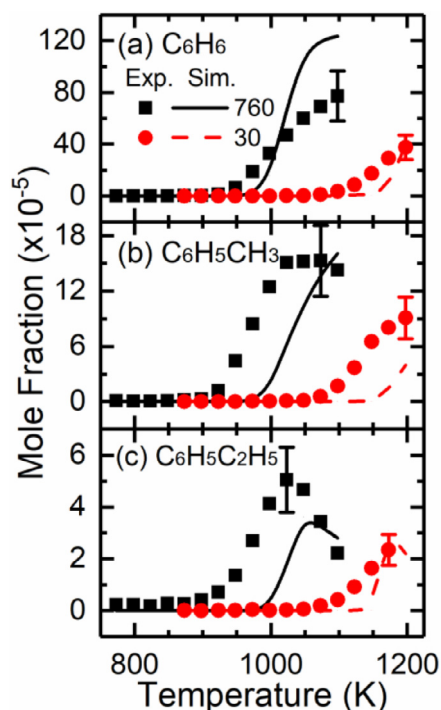


Fig. 3. Experimental and modeling results of aromatic intermediates during pyrolysis of T124MCH.

estimated from other mechanisms where errors still remain. On the other hand, although aromatic hydrocarbons were attractive intermediates in combustion, oxidation or pyrolysis of alkylcyclohexanes, it is difficult to illustrate their formation mechanisms and to predict accurate mole fraction values.

4.2. ROP analysis

To reveal the detailed process of T124MCH pyrolysis, flux and sensitivity analysis were performed at 30 Torr (59.5% at 1150 K) and 760 Torr (52.2% at 975 K). The ROP analysis results of T124MCH pyrolysis under two pressures are shown in Figs. 4–6. The pyrolysis consumption pathways after dehydrogenation on the primary carbons are shown in Fig. 4. To reveal the effect

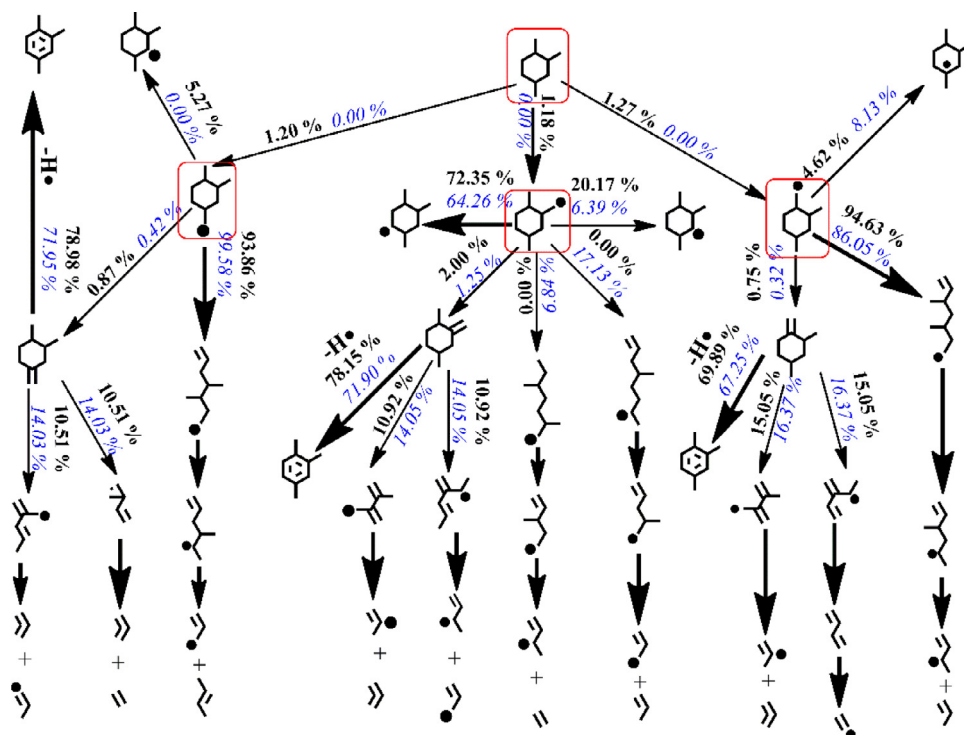


Fig. 4. Reaction flux analysis of 1,2,4-trimethylcyclohexane pyrolysis, H-abstraction from the primary carbons (bold – $p = 760$ Torr, $T = 975$ K, italic – $p = 30$ Torr, $T = 1150$ K).

of equivalence ratio, the corresponding pathways under oxidation conditions are presented in Fig. S1 (SM). Only a small amount of T124MCH (3.6%) was consumed through dehydrogenation on the primary carbon atoms during pyrolysis, far less than the counterpart in oxidation (15–20%), suggesting that dehydrogenation on the primary carbon atoms would rely on reactions between T124MCH molecule and oxidized radicals (like OH and HO₂) or oxygen molecule, which could not continue without oxygen. At the same time, the absence of oxygen will make a difference in following reactions of methyl dehydrogenation radicals, but the main reaction pathways would not change obviously. Variation of pressure has a slight effect on reaction pathways, especially on secondary or further reactions. Since many reactions could hardly proceed for lack of oxygen atom in pyrolysis, the branching ratio of each consumption pathway would not change even if pressure caused effects on reaction rates. Another reason is that the pressure effect was only considered in initial and subsequent decomposition reactions when constructing the kinetic model, and the pressure effects on secondary dehydrogenation were ignored. The rate constants of dehydrogenation reactions on the secondary carbon atoms mainly come from estimation of the mechanism of MCH, with limited precision. Precise quantification calculation of these reactions need to be explored in future work.

Pyrolysis consumption pathway of dehydrogenated radicals on three tertiary carbon positions in T124MCH is displayed in Fig. 5. Compared to Fig. S2 (SM), the percentage of T124MCH consumed in these ways varies slightly, which means these reactions are not main reaction pathways in both pyrolysis and oxidation. The reason is that it is difficult to continue the dehydrogenation at the position where a methyl group has replaced by a hydrogen atom, since the barrier of dehydrogenation reaction on tertiary carbon is much higher than primary or secondary carbon. Partial reactions terminated or proceeded on opposite direction under pyrolysis condition, relative to vacant oxygen atom. This indicates that oxygen content has more profound effect than pressure, which can be seen from the results shown in Figs. S1–S3.

The consumption pathways of radicals dehydrogenated from secondary carbons of T124MCH is presented in Fig. 6. It is obvious that T124MCH tends to decompose through dehydrogenation in secondary carbon atoms under pyrolysis condition compared to that in oxidation (see Fig. S3 in SM). About 80% at 760 Torr and 90% at 30 Torr of T124MCH consumes via this pathway at pyrolysis, much higher than 30–40% at oxidation. As most T124MCH dehydrogenates through unimolecular decomposition in pyrolysis instead of H-abstraction with radicals in oxidation, the real changes in these ways become obvious. The dominant consumption pathway of T124MCH is not unimolecular decomposition due to the existence of various radicals under oxidation. Another difference between pyrolysis and oxidation is that, under pyrolysis condition, the main consumption channel of radicals formed by H-abstraction of T124MCH is no longer through ring opening isomerization and dehydrogenation, but tends to proceed with the ipso-additions such as methyl group abstraction with more radicals to form a benzene ring. Benzene ring will further convert to toluene and ethylbenzene, explaining the emergence of large amount of aromatic intermediates in Fig. 3.

4.3. ROP and sensitivity analysis

Sensitivity analysis of 124MCH pyrolysis under two pressures is shown in Fig. 7. At 760 Torr, the most inhibiting reactions are direct isomerization of T124MCH, since the products of these reactions, such as C₈H₁₅-5-5M and C₇H₁₂-2-56M, will consume radicals like H and CH₃ in their following reactions. There is no reaction that will obviously inhibit the consumption of T124MCH at 30 Torr. The most promoting reactions for the decomposition of T124MCH are related to the formation or consumption of H₂ and CH₄ in 760 Torr. The formation of CH₃ is the most promoting reaction that enhances the pyrolysis of T124MCH, such as T124MCH = S1XD24MCH + CH₃ and T124MCH = S4XD12MCH + CH₃ at 30 Torr, CH₄ + H = CH₃ + H₂ at 760 Torr, indicating that CH₃ plays an important role in the

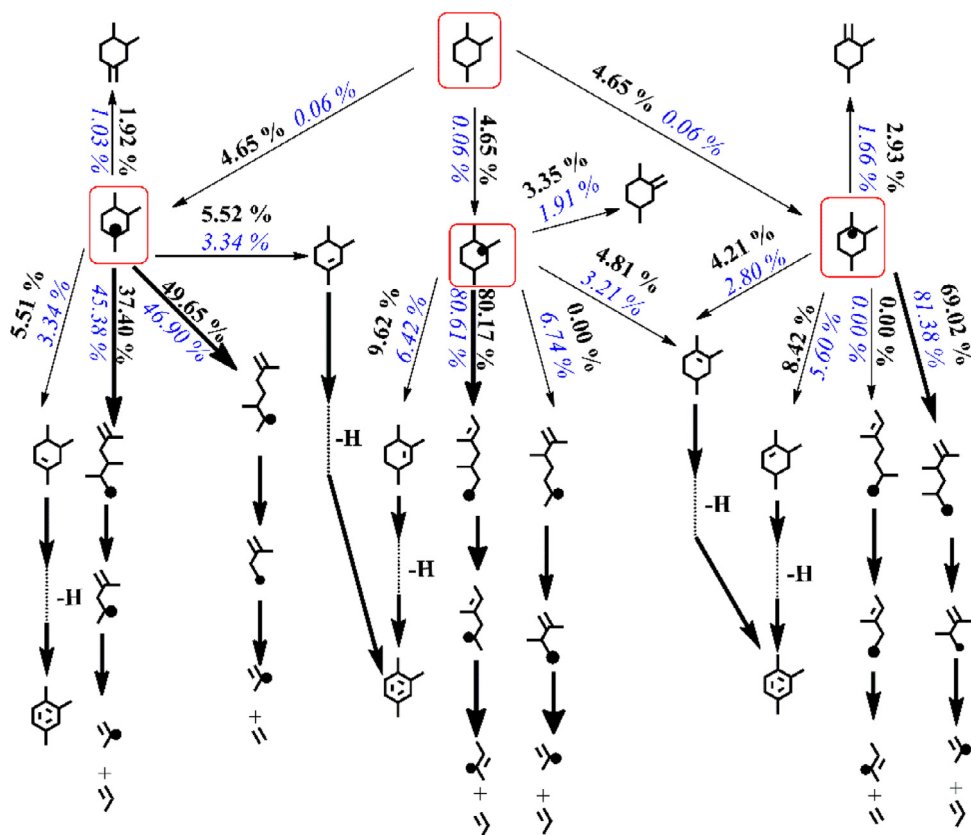


Fig. 5. Reaction flux analysis of 1,2,4-trimethylcyclohexane pyrolysis, H-abstraction from the tertiary carbons (bold – $p = 760$ Torr, $T = 975$ K, italic – $p = 30$ Torr, $T = 1150$ K).

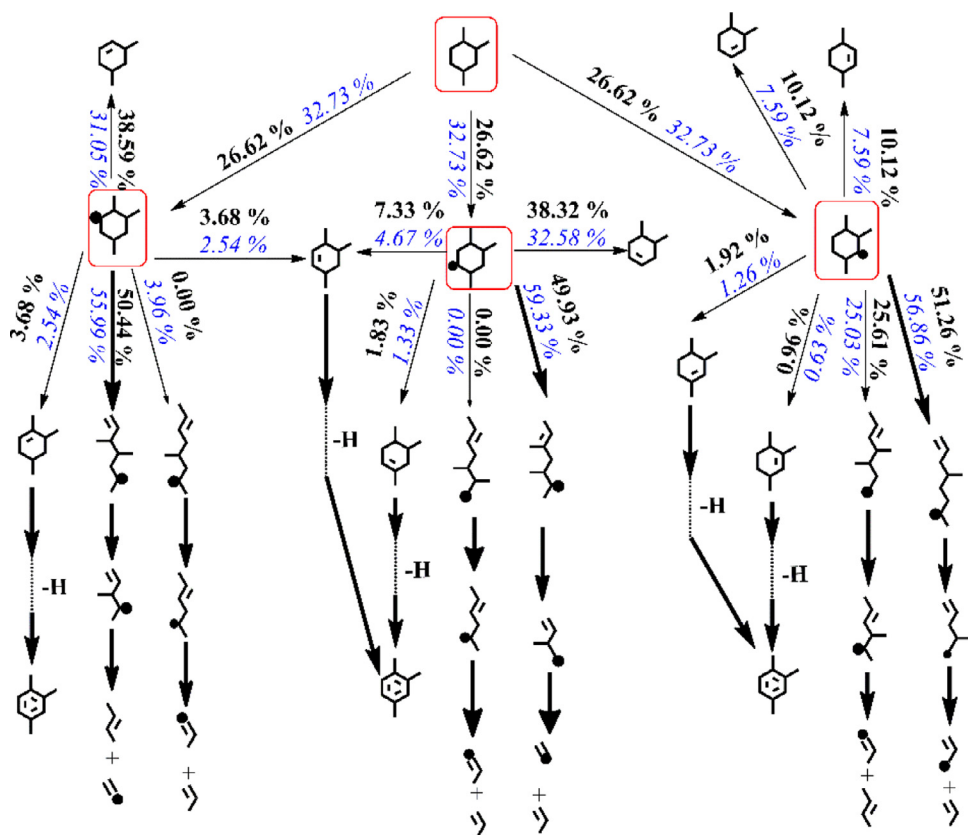


Fig. 6. Reaction flux analysis of 1,2,4-trimethylcyclohexane oxidation, H-abstraction from the secondary carbons (bold – $p = 760$ Torr, $T = 975$ K, italic – $p = 30$ Torr, $T = 1150$ K).

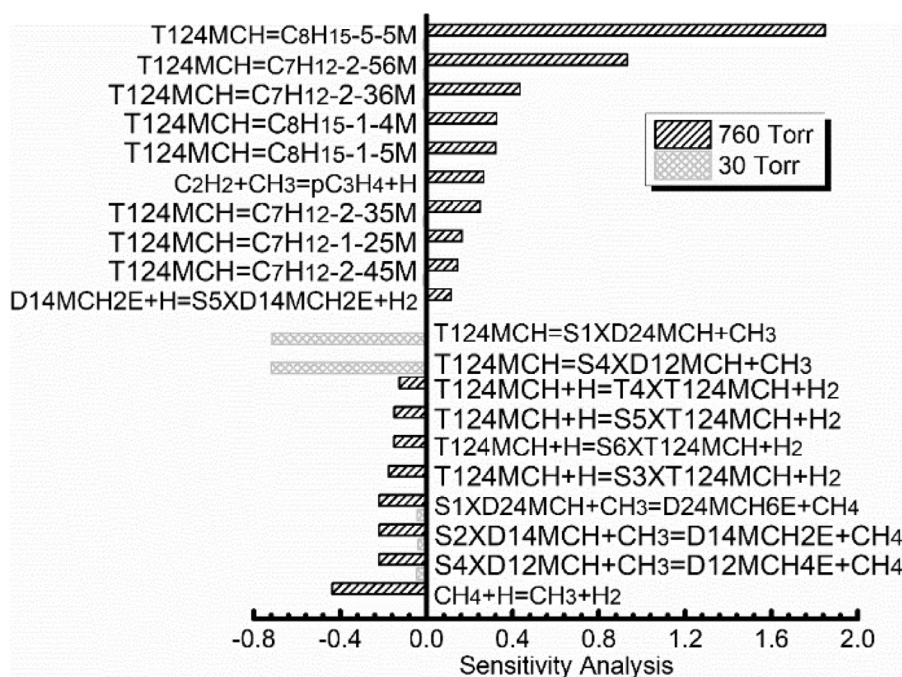


Fig. 7. Sensitivity analysis results of 1,2,4-trimethylcyclohexane pyrolysis under 760 and 30 Torr.

pyrolysis of T124MCH. It should be noted that the H and OH radicals play similarly significant roles in the oxidation of T124MCH [19].

4.4. Comparison of aromatics formation for T124MCH under oxidation and pyrolysis conditions

From the sensitivity analysis results shown in Fig. 7, the formation of aromatics has a significant inhibiting effect on the consumption of T124MCH in pyrolysis. To find the influence of pressure and equivalence ratio on the production of aromatic hydrocarbons, comparison was performed among pyrolysis and oxidation results, as shown in Fig. 8. In order to eliminate the effect of different import concentration of fuel, the concentration of all intermediates was normalized (divided by the fuel inlet concentration) and defined as normalized mole fraction. The oxidation results come from [19].

With the increase of equivalence ratio, the normalized mole fractions of benzene and ethylbenzene increase under the same pressure. Benzene was mainly produced by the reactions between C₃H₃ and C₃H₄ at $\phi = 0.4$. There are more C₃H₃ and C₃H₄ in rich oxidation and pyrolysis conditions, leading to a higher forming rate under rich oxidation or pyrolysis conditions. Besides, there are other pathways producing benzene, such as H-abstraction from C₅H₅CH₂ radicals. The only pathway to form ethylbenzene is the reaction between benzyl and methyl group at both oxidation and pyrolysis conditions. Most ethylbenzene is consumed through hydrogen abstraction with methyl group or hydrogen atom under pyrolysis condition. With the increase of oxygen, H-abstraction reaction between ethylbenzene and hydroxyl or peroxide hydroxyl consumes more ethylbenzene, leading to less amount of ethylbenzene.

The production of toluene is relatively stable, irrelevant of equivalence ratios. There are two pathways forming toluene. It can be formed through H-abstraction from a C₇H₉ radical, which plays an important role in oxidation condition, or the combination between C₃H₃ and C₃H₄, which is the main way at pyrolysis condition. The two reactions compensate with each other, leading to a stable normalized mole fraction of toluene. The formation of aro-

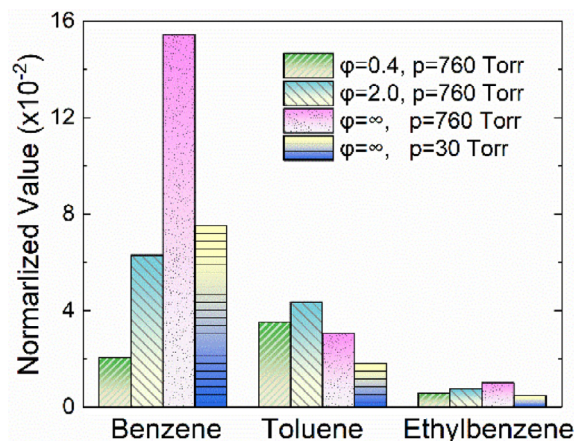


Fig. 8. Comparison of aromatics between pyrolysis and oxidation of T124MCH.

omatics increases generally as the equivalence ratio increases. The reason is that benzene can be consumed by many routes with the existence of oxygen atom or oxidized radicals. The abundant aromatics under pyrolysis conditions will combine with radicals to form T124MCH, and the reversible reactions could inhibit the consumption of T124MCH under pyrolysis condition.

The three aromatics are easier to be formed during the pyrolysis of T124MCH at higher pressure. C₃H₃ and aC₃H₄ are responsible for the formation of benzene at 760 Torr. H-abstraction on C₅H₅CH₂ also contributes to benzene at ambient pressure. Phenyl group, a radical mainly comes from benzene, plays an important role in the formation of benzene at 30 Torr. The reaction between C₃H₃ and pC₃H₄ makes a minor contribution to benzene formation at low pressure. The slower net rate makes a lower concentration for benzene at 30 Torr. Reaction between C₃H₃ and C₄H₆ plays a dominant role in toluene formation at atmospheric pressure, while phenyl radical takes into a little percentage for the production of toluene. At low pressure, more than 90% of toluene is formed via

$A1- + CH_3 (+ M) = A1CH_3 (+ M)$, which means the formation of toluene is limited to benzene and phenyl radical. Most of ethylbenzene comes from the reaction between methyl and benzyl group at both pressures, and the abundance of benzyl radical makes a higher rate at 760 Torr.

5. Conclusions

This work reports the pyrolysis study of T124MCH in a flow reactor equipped with an SVUV-photoionization molecular-beam mass spectrometry at 30 and 760 Torr. The results indicate that the pyrolysis of T124MCH is significantly influenced by the pressure. The onset temperature of decomposition decreases from 1050 to 875 K as the pressure increases from 30 to 760 Torr. The main light hydrocarbons and aromatics increase monotonically with temperature at 30 Torr, and have a peak value at 760 Torr. Propyne, 1,3-butadiene and 2-butene have more yields at lower pressure. The value of 2-butylene is almost equal at both pressures. The other species will be more formed at 760 Torr. According to the ROP analysis, T124MCH consumes mainly through the H-abstraction reactions taking place on C6 ring at low pressure, which is less important at atmospheric condition. The sensitivity analysis shows that the isomerization reactions of T124MCH yielding to branch-chain alkyls will significantly inhibit its consumption at 760 Torr, while the abstraction of methyl on its derived radicals will promote the pyrolysis of T124MCH. At 30 Torr, there is no reaction obviously inhibiting the pyrolysis process, and the methyl abstraction on T124MCH is the most promoting reaction. The species profiles of pyrolysis at 760 Torr are approximate to rich oxidation, apart from the difference that abundant 1,3-butadiene was observed in pyrolysis. Compared to the oxidation pathway, more T124MCH will decompose through H-abstraction on secondary instead of primary carbon. CH_3 radical plays the most important role in T124MCH pyrolysis, as important as H and OH in oxidation. These results could enrich the products pool of T124MCH pyrolysis and help in further understanding the combustion characteristic of T124MCH.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work. There is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2020.06.020](https://doi.org/10.1016/j.combustflame.2020.06.020).

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